

Tracing-KalmanNet: Deep Kalman Filtering for Nonlinear and Intermittent Inertial Data

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Abstract

Kalman filters are fundamental to attitude estimation and inertial navigation, effectively mitigating sensor errors and drift to enhance the accuracy and stability of systems. However, their performance degrades significantly in complex scenarios characterized by nonlinear dynamics, intermittent observations, and unknown noise statistics. To address these challenges, we propose Tracing-KalmanNet (TKN), a deep learning-enhanced filtering framework that integrates neural networks into the Kalman gain computation. Unlike traditional methods, TKN operates without prior knowledge of noise distributions or covariance models, achieving robust performance in highly nonlinear and data-sparse environments. Moreover, by leveraging the temporal context of historical inertial data, TKN dynamically adapts its parameters to improve generalization and resilience to model mismatch. Extensive experiments demonstrate that TKN consistently outperforms classical and learned Kalman filter variants under discontinuous and nonlinear inertial measurements, offering a promising solution for attitude estimation in degraded sensing conditions.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

Keywords

deep Kalman filtering, attitude estimation, nonlinear dynamics, intermittent observations

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1 Introduction

The inertial measurement unit (IMU) estimates motion attitude using only accelerometers and gyroscopes, without external references. As attitude estimation becomes increasingly vital in aerospace [22], robotics [20], and autonomous systems [21], ensuring robustness under low sampling rates and noisy conditions remains a fundamental challenge [24].

Kalman filtering provides a solution by iteratively updating state estimates based on system observations [5, 6, 26]. Its low computational complexity, strong theoretical foundation, and excellent performance under known noise distributions have made it widely adopted in navigation positioning [18], attitude perception [7] and 3D tracking [23], forming a cornerstone for tackling the aforementioned challenges [16]. To enhance its applicability, several nonlinear variants have been developed, including the extended Kalman filter (EKF) [25], unscented Kalman filter (UKF) [14], particle filter (PF) [3, 4], among others. However, in practice, IMU often operates under complex nonlinear dynamics, sparse and asynchronous observations, and unknown or time-varying noise characteristics. These conditions hinder accurate modeling and parameter identification [11], resulting in degraded performance and limited generalization of traditional model-based filters in real-world scenarios.

The advent of deep learning offers new opportunities to address the limitations of traditional filtering methods [1, 9]. As research into data-driven deep learning algorithms advances, systems can now design controllers and learn estimation functions directly from output data or extracted features, eliminating the need for precise mathematical models of the controlled object [8, 17]. KalmanNet exemplifies this trend by embedding recurrent neural networks (RNNs) into the Kalman filtering framework to learn Kalman gains from data while preserving the filter's recursive structure [15]. This

approach retains the low complexity and stability of traditional Kalman Filters while enhancing adaptability to unknown system dynamics, significantly improving estimation accuracy and robustness. However, existing KalmanNet architectures struggle to handle low-frequency or intermittent measurements and fail to fully leverage temporal dependencies, leading to suboptimal performance in scenarios with sparse or noisy observations [2].

To overcome these limitations, this study proposes an innovative deep learning-enhanced recursive filtering framework, Tracing-KalmanNet (TKN). TKN embeds a Transformer into the EKF framework to more effectively capture long-range temporal dependencies. This design enables data-driven Kalman gain estimation, eliminating the reliance on prior noise statistics and the need for iterative linearization inherent in classical EKF. By incorporating historical motion data into its input, TKN infers latent temporal dynamics and adaptively responds to state variations, thereby achieving enhanced generalization and robustness under perceptual degradation. We evaluated TKN in a real-world IMU task using a radiosonde platform [10]. Due to platform limitations, the IMU data exhibit low transmission frequency, strong nonlinearity, and intermittent sensor updates, which are conditions under which classical filters often fail. Experimental results demonstrate that TKN outperforms existing filtering methods, achieves the lowest mean squared error (MSE) in attitude estimation, and maintains high robustness under adverse measurement conditions. The contributions of this paper are as follows:

- **Neural Kalman Gain Learning:** TKN integrates neural networks into the Kalman gain computation, removing the dependence on prior noise characteristics and enabling robust performance under unknown and time-varying noise conditions.
- **Time-Aware Architecture:** By incorporating historical motion information into the filter's input, TKN enhances adaptability to non-stationary dynamics and improves generalization across unseen trajectories.
- **Robustness in Intermittent Observation Scenarios:** Extensive evaluations on radiosonde-based IMU data demonstrate that TKN surpasses existing Kalman-based methods in accuracy and stability under nonlinear dynamics and sparse observation conditions.

2 Tracing-KalmanNet

In the classical EKF, attitude estimation for the IMU is performed using the gyroscope's angular velocity as the state transition input and the accelerometer's output as the observation. The accuracy of the estimation primarily depends on the Kalman gain, which balances model prediction with measurement correction. The Kalman gain κ_t is typically computed as follows:

$$\begin{cases} \kappa_t = P_t \cdot H_t^T \cdot (H_t \cdot P_t \cdot H_t^T + R)^{-1} \\ P_t = F_t \cdot P_{t-1} \cdot F_t^T + Q \end{cases}, \quad (1)$$

$$Q = q^2 \cdot I, \quad R = r^2 \cdot I, \quad v \triangleq \frac{q^2}{r^2}, \quad (2)$$

where P_t denotes the covariance matrix of the state vector, H_t is the observation matrix, and F_t is the state transition matrix. R and Q

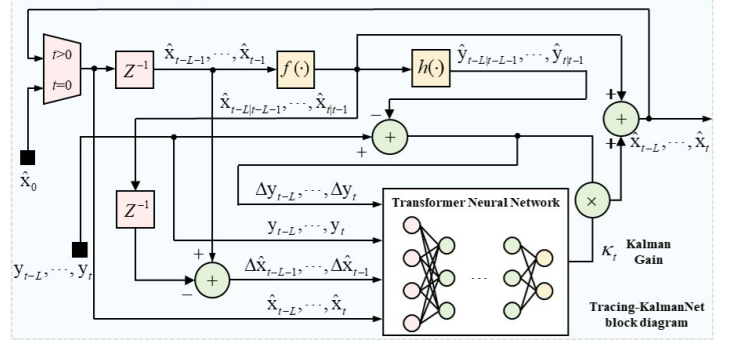


Figure 1: Tracing-KalmanNet block diagram.

represent the observation noise covariance and process noise covariance, respectively. This formulation assumes that noise statistics are known and Gaussian, and that the system model is accurately defined. However, in practical IMU applications, these assumptions are rarely valid. Sensor noise often exhibits non-Gaussian and time-varying behavior, and the system dynamics are typically nonlinear and difficult to model. Moreover, data transmission may be intermittent and low-frequency, leading to discontinuities in the available observations.

To address these challenges, this study proposes Tracing-KalmanNet (TKN), a data-driven filtering framework that replaces the analytical computation of the Kalman gain with a neural network predictor. By learning to infer the Kalman gain directly from observed data, TKN eliminates the need for prior knowledge of noise distributions and system dynamics, thereby enhancing robustness under nonlinear dynamics and intermittent observation conditions.

Unlike traditional filters that compute the Kalman gain based on a fixed system matrix, TKN employs a neural network to infer the optimal gain from the statistical and temporal characteristics of sensor data. To achieve this, the network must be provided with input features that encapsulate the information necessary to estimate the Kalman gain. In attitude estimation tasks, uncertainty significantly affects algorithm performance. Therefore, in addition to including statistical inputs such as the current observation y_t and the prior state estimate $\hat{x}_{t-1}$ at each time step L , TKN also innovatively tracks historical attitude data obtained from the accelerometer and gyroscope. Specifically, the historical data provide the network with contextual insight, enabling it to capture both long-term trends and short-term fluctuations in system dynamics. This allows TKN to dynamically adjust model parameters, adapt to variations in the data, and enhance predictive and control performance. The architecture block diagram of TKN is presented in Fig 1, and the corresponding input features include:

- Observation difference: $\{\Delta y_{t-L}, \Delta y_{t-L+1}, \dots, \Delta y_t\}$,
- State update difference: $\{\Delta \hat{x}_{t-L-1}, \Delta \hat{x}_{t-L}, \dots, \Delta \hat{x}_{t-1}\}$,
- Historical observation data: $\{y_{t-L}, y_{t-L+1}, \dots, y_t\}$,
- Historical state estimation data: $\{\hat{x}_{t-L}, \hat{x}_{t-L+1}, \dots, \hat{x}_t\}$.

Where the observation difference is defined as $\Delta y_t = y_t - \hat{y}_{t|t-1}$ and the state update difference as $\Delta \hat{x}_t = \hat{x}_{t|t} - \hat{x}_{t|t-1}$. These inputs provide the network with both instantaneous statistical deviations and long-term trends in system evolution. The residual terms help

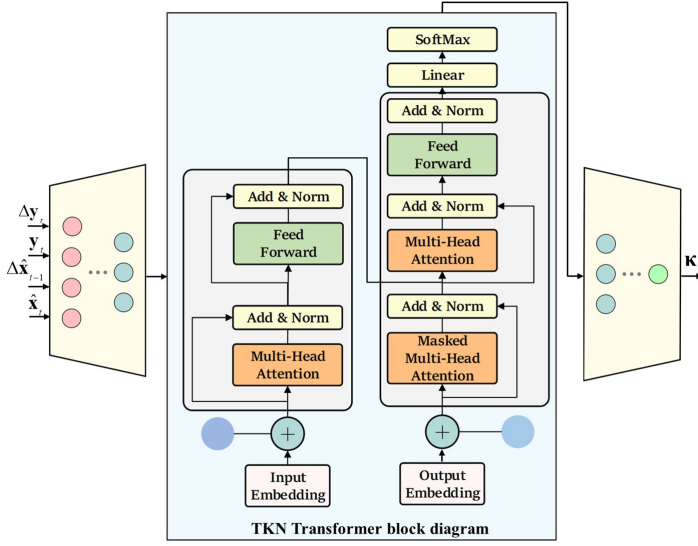


Figure 2: TKN Transformer block diagram.

the model focus on unpredictable components in the state and observation data, which are highly influenced by noise characteristics. Additionally, the inclusion of historical data enables the model to capture temporal dependencies, identify patterns of measurement dropout, and adapt to dynamic changes in the noise distribution.

Regarding the core network architecture, the conventional Kalman Net typically employs RNNs to process sequential inputs. However, RNNs inherently struggle to capture long-range dependencies and are prone to issues such as vanishing gradients in longer sequences. To overcome these limitations, TKN adopts a Transformer-based architecture that leverages self-attention mechanisms to model pairwise interactions across the entire input sequence, effectively capturing both short-term fluctuations and long-term trends in sensor behavior [19]. Furthermore, by applying multi-head attention to both difference terms and historical features, TKN is able to learn time-varying noise patterns. The architecture also supports parallel computation, accelerates training convergence, and offers greater flexibility compared to traditional models such as RNNs [12, 13]. As illustrated in Fig 2, the TKN Transformer architecture comprises a fully connected input layer, a Transformer encoder layer, and a final output layer that predicts the Kalman gain. The number of attention heads, hidden layer dimensions, and network depth are configurable as hyperparameters.

TKN employs a supervised learning neural network to predict the Kalman gain in an end-to-end manner. During backpropagation, the loss function is propagated through the Kalman update equations, allowing gradients to be effectively transmitted to the Kalman gain prediction module. The specific loss function is defined as follows:

$$\ell = \|\mathbf{x}_t - \hat{\mathbf{x}}_{t|t}\|^2 = \|\kappa_t \Delta \mathbf{y}_t - \Delta \mathbf{x}_t\|^2, \quad (3)$$

where $\Delta x_t = x_t - \hat{x}_{t|t-1}$. This formulation enables the loss function to be backpropagated to the Kalman gain, allowing the gradient of the loss with respect to the Kalman gain to be directly computed from the output of TKN:

$$\frac{\partial \ell}{\partial \kappa_t} = \frac{\partial \|\kappa_t \Delta y_t - \Delta x_t\|^2}{\partial \kappa_t} = 2 \cdot (\kappa_t \Delta y_t - \Delta x_t) \cdot \Delta y_t^T. \quad (4)$$

Based on the above process, TKN avoids reliance on externally provided true Kalman gains by learning to approximate the optimal gain solely from historical estimation errors and observation differences. This enables the system to effectively adapt to environments characterized by unknown, non-stationary, and non-Gaussian noise.

3 Evaluation

3.1 Simulation Scenario Evaluation

To comprehensively evaluate the performance of TKN, we simulated the attitude estimation task under three representative scenarios that reflect common challenges encountered in real-world IMU, including model mismatch, dynamic uncertainty, and non-ideal noise distributions:

A. Full State and Known Noise Statistics

Complete state space information enables accurate modeling of the error covariance matrix in the classic EKF, thereby allowing it to achieve optimal estimation performance. This scenario serves as a theoretical upper bound by simulating a system with a known noise covariance matrix. By varying the noise ratio to represent different noise conditions, the experiment evaluates whether TKN can match the performance of model-based filters under ideal circumstances.

B. Randomized Noise Covariances

In practical applications, full state knowledge is rarely available. To emulate real-world model uncertainty, the noise covariance is randomly perturbed during both training and inference. This scenario evaluates the robustness of each filtering method to model mismatch and parameter deviations, testing TKN’s ability to correct for random estimation errors.

C. Non-Gaussian Noise Distributions

Real sensor data often exhibit skewed or heavy-tailed noise characteristics, such as those from Laplace or mixed Gaussian distributions, which deviate from the Gaussian assumption. This scenario assesses the adaptability of TKN and baseline filters when the noise distribution is non-Gaussian.

We evaluate TKN’s capability to learn system error characteristics by comparing the MSE of attitude estimation with benchmark algorithms across these scenarios, as shown in Table. 1.

The results demonstrate that TKN consistently achieves the lowest MSE across all test conditions. Under fully observable and ideal circumstances, its performance aligns with that of model-based filters, indicating that its data-driven Kalman gain estimation does not compromise accuracy. As scenario complexity increases, TKN significantly outperforms all baselines, showcasing strong robustness to model mismatches and distribution shifts. This superior performance is attributed to TKN’s ability to implicitly learn complex noise characteristics and dynamic system behaviors directly from data, without relying on explicit prior models. By leveraging long-range temporal dependencies through its internal architecture, TKN dynamically adapts its parameters, achieving robust and generalizable estimation performance under diverse operating conditions.

Table 1: MSE (dB) Comparison of TKN and Benchmark Algorithms Across Scenarios.

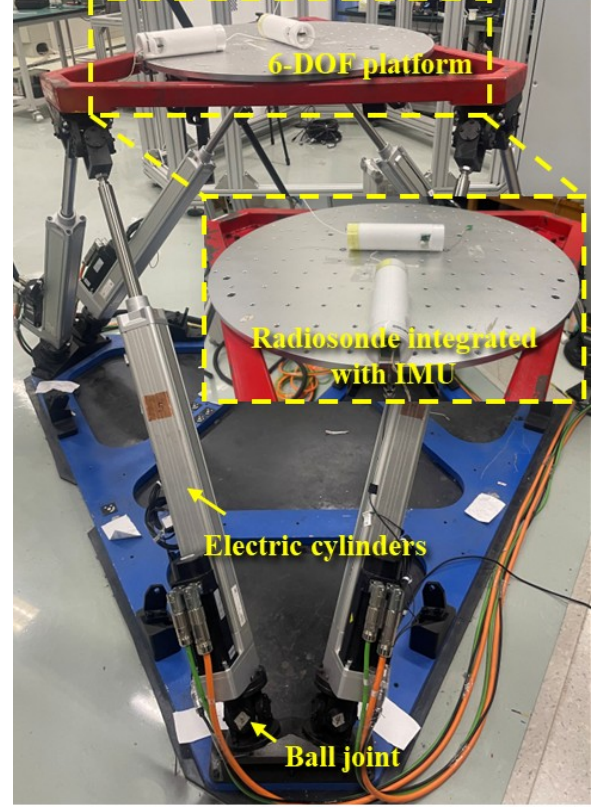
Scenario	ν (dB)	EKF	PF	KalmanNet	TKN
A	-20	-8.19	-7.75	-7.57	-7.92
	-10	-9.35	-9.00	-9.18	-9.32
	0	-18.29	-17.33	-17.63	-18.10
	10	-22.56	-21.68	-21.55	-22.34
	20	-30.45	-28.35	-29.91	-29.96
B	-20	-3.60	-7.43	-7.27	-8.58
	-10	-6.34	-8.78	-9.09	-9.01
	0	-14.75	-17.19	-17.50	-18.19
	10	-20.83	-21.50	-21.44	-22.06
	20	-23.79	-28.24	-29.77	-29.82
C	-20	-4.54	-4.34	-5.26	-7.99
	-10	-6.46	-6.86	-9.97	-12.06
	0	-14.34	-14.96	-13.12	-16.62
	10	-15.28	-18.36	-19.90	-22.97
	20	-16.78	-19.42	-20.17	-24.97

3.2 Practical Evaluation on Radiosonde IMU with Intermittent Observations

To further evaluate the practical performance of TKN, we selected a radiosonde system integrated with an IMU as the test subject and conducted experiments on a six-degree-of-freedom (6-DOF) vibration test platform, as illustrated in Fig 3. This platform comprises six electric actuators, twelve ball joints, and two stacked platforms. The lower platform remains stationary, while the upper platform moves freely along 6-DOF, effectively simulating the complex flight dynamics of the radiosonde.

In real-world scenarios, the radiosonde's IMU operates under strict power constraints and limited data transmission bandwidth, allowing attitude data transmission at only 1 Hz. This low-frequency and intermittent observation regime, combined with highly nonlinear environmental conditions, presents significant challenges for conventional filtering methods in maintaining accurate and stable attitude estimates. To replicate these operational conditions, we simulated various modes by adjusting the platform's motion angles and frequencies. Raw IMU data and ground truth attitude data from the 6-DOF system were used to train and evaluate the network under these complex, sparse-input conditions.

By varying the coning motion angle, we constructed datasets representing different dynamic behaviors. The resulting MSE of attitude estimation, processed using TKN and other baseline algorithms, is presented in Fig 4 and detailed in Table 2. Results indicate that TKN consistently achieves the lowest MSE across all scenarios, effectively addressing the challenges posed by intermittent measurements and nonlinear dynamics. Unlike traditional filters that rely on continuous observations or static noise models, TKN dynamically captures temporal dependencies from historical data and adapts the Kalman gain accordingly, leading to superior estimation accuracy and generalization performance. Compared with KalmanNet, TKN reduces the average MSE by 6.64 dB, highlighting its advantage in managing sparse and noisy inputs.

**Figure 3: Attitude Estimation Experimental Setup Using a 6-DOF Vibration Platform.****Table 2: MSE (dB) of different methods.**

Angel (°)	AD	AVI	EKF	KalmanNet	TKN
5	-17.53	-17.71	-18.50	-19.57	-27.77
8	-1.32	-7.09	-11.48	-12.11	-18.90
10	10.06	2.53	2.32	2.11	-3.42
15	22.71	24.09	15.80	12.22	6.19

To further demonstrate the effectiveness of TKN, we visualize the estimated motion trajectories generated by various algorithms in Fig 5. TKN's trajectory closely matches the ground truth, exhibiting improved smoothness, accuracy, and robustness. These results confirm TKN's superiority in handling data sparsity and complex noise, supporting its practical deployment in radiosonde applications.

4 Conclusion

This study proposes Tracing-KalmanNet (TKN), a novel deep learning-based filtering framework that integrates Transformer networks with the EKF. By learning Kalman gains from historical inertial sequences, TKN eliminates the need for prior noise statistics and improves adaptability to model mismatches and non-Gaussian disturbances. The method delivers robust and accurate attitude estimation in both simulated environments and real-world radiosonde

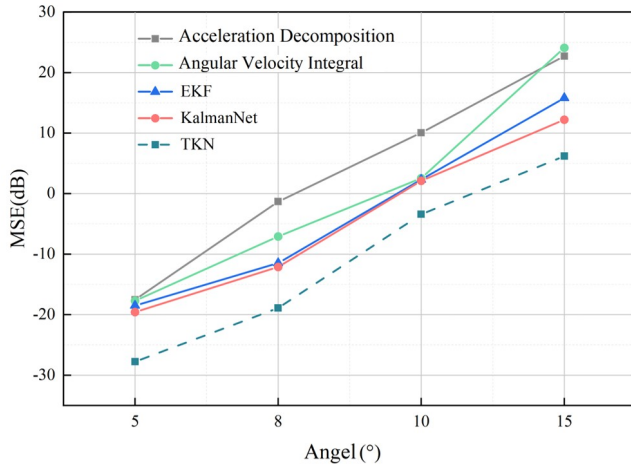


Figure 4: Attitude Estimation MSE Across Different Methods and Motion Angles.

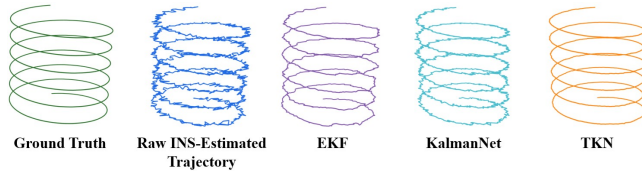


Figure 5: Visualization of Posture Estimation Results Across Algorithms.

IMU experiments, significantly outperforming classical filtering algorithms, particularly under intermittent observations and complex dynamic conditions. These results highlight the potential of TKN as a generalizable and practical solution for robust state estimation in real-world IMU applications. Future research will focus on real-time deployment and extension to multi-sensor fusion scenarios.

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References

- [1] Yoshua Bengio et al. 2009. Learning deep architectures for AI. *Foundations and trends® in Machine Learning* 2, 1 (2009), 1–127.
- [2] Itay Buchnik, Guy Revach, Damiano Steger, Ruud JG Van Sloun, Tirza Routtenberg, and Nir Shlezinger. 2023. Latent-KalmanNet: Learned Kalman filtering for tracking from high-dimensional signals. *IEEE Transactions on Signal Processing* (2023).
- [3] Xinlei Chen, Aveek Purohit, Carlos Ruiz Dominguez, Stefano Carpin, and Pei Zhang. 2015. Drunkwalk: Collaborative and adaptive planning for navigation of micro-aerial sensor swarms. In *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems*. 295–308.
- [4] Xinlei Chen, Carlos Ruiz, Sihan Zeng, Liyao Gao, Aveek Purohit, Stefano Carpin, and Pei Zhang. 2020. H-DrunkWalk: Collaborative and adaptive navigation for heterogeneous MAV swarm. *ACM Transactions on Sensor Networks (TOSN)* 16, 2 (2020), 1–27.
- [5] Xinlei Chen, Susu Xu, Xinyu Liu, Xiangxiang Xu, Hae Young Noh, Lin Zhang, and Pei Zhang. 2020. Adaptive hybrid model-enabled sensing system (HMSS) for mobile fine-grained air pollution estimation. *IEEE Transactions on Mobile Computing* 21, 6 (2020), 1927–1944.
- [6] Xinlei Chen, Xiangxiang Xu, Xinyu Liu, Shijia Pan, Jiayou He, Hae Young Noh, Lin Zhang, and Pei Zhang. 2018. Pga: Physics guided and adaptive approach for mobile fine-grained air pollution estimation. In *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers*. 1321–1330.
- [7] Geon Choi, Jeonghun Park, Nir Shlezinger, Yonina C Eldar, and Namyoon Lee. 2023. Split-KalmanNet: A robust model-based deep learning approach for state estimation. *IEEE Transactions on Vehicular Technology* 72, 9 (2023), 12326–12331.
- [8] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural computation* 9, 8 (1997), 1735–1780.
- [9] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. 2015. Deep learning. *nature* 521, 7553 (2015), 436–444.
- [10] Dapeng Li, Aobei Chen, Xiaoyuan Dong, Dezhi Zheng, and Rui Na. 2025. Modeling analysis and experimental verification of IDE humidity sensor integrated with PZT: Ultra-fast response and recovery times. *Sensors and Actuators B: Chemical* 426 (2025), 136985.
- [11] Dapeng Li, Aobei Chen, Rui Na, Jiaqi Jing, Chun Hu, Dezhi Zheng, and Shuai Wang. 2023. Microstructure design and optimization of high-sensitivity interdigital capacitive humidity sensor based on uncertainty analysis. *Measurement* 222 (2023), 113599.
- [12] Dapeng Li, Hechu Zhang, Aobei Chen, Xiaoyuan Dong, Yu Yang, Dezhi Zheng, and Rui Na. 2024. Adaptive temperature compensation for MoS₂ humidity sensor in complex environments using ISSA-BP neural network. *Sensors and Actuators A: Physical* 379 (2024), 115982.
- [13] Dapeng Li, Hechu Zhang, Yu Yang, Wei Li, Shuai Wang, and Dezhi Zheng. 2024. Temperature compensation for humidity sensors using ISSA-BP neural network. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*. 2004–2009.
- [14] Wangyan Li, Guoliang Wei, Fei Han, and Yurong Liu. 2015. Weighted average consensus-based unscented Kalman filtering. *IEEE transactions on cybernetics* 46, 2 (2015), 558–567.
- [15] Guy Revach, Nir Shlezinger, Xiaoyong Ni, Adria Lopez Escoriza, Ruud JG Van Sloun, and Yonina C Eldar. 2022. KalmanNet: Neural network aided Kalman filtering for partially known dynamics. *IEEE Transactions on Signal Processing* 70 (2022), 1532–1547.
- [16] Chong Shen, Yu Zhang, Xiaoting Guo, Xiyuan Chen, Huiliang Cao, Jun Tang, Jie Li, and Jun Liu. 2020. Seamless GPS/inertial navigation system based on self-learning square-root cubature Kalman filter. *IEEE Transactions on Industrial Electronics* 68, 1 (2020), 499–508.
- [17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).
- [18] Haoyang Wang, Xuecheng Chen, Yuhua Cheng, Chenye Wu, Fan Dang, and Xinlei Chen. 2022. H-swarmloc: Efficient scheduling for localization of heterogeneous mav swarm with deep reinforcement learning. In *Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems*. 1148–1154.
- [19] Haoyang Wang, Jingao Xu, Chenyu Zhao, Zihong Lu, Yuhua Cheng, Xuecheng Chen, Xiao-Ping Zhang, Yunhao Liu, and Xinlei Chen. 2024. Transformloc: Transforming mavs into mobile localization infrastructures in heterogeneous swarms. In *IEEE INFOCOM 2024-IEEE Conference on Computer Communications*. IEEE, 1101–1110.
- [20] Xin Xia, Neel P Bhatt, Amir Khajepour, and Ehsan Hashemi. 2023. Integrated inertial-LiDAR-based map matching localization for varying environments. *IEEE transactions on intelligent vehicles* (2023).
- [21] Xiaobin Xu, Lei Zhang, Jian Yang, Chenfei Cao, Wen Wang, Yingying Ran, Zhiying Tan, and Minzhou Luo. 2022. A review of multi-sensor fusion slam systems based on 3D LIDAR. *Remote Sensing* 14, 12 (2022), 2835.
- [22] Yuanxi Yang, Yue Mao, and Bijiao Sun. 2020. Basic performance and future developments of BeiDou global navigation satellite system. *Satellite navigation* 1, 1 (2020), 1.
- [23] Jirong Zha, Yuxuan Fan, Kai Li, Han Li, Chen Gao, Xinlei Chen, and Yong Li. 2025. DIMM: Decoupled Multi-hierarchy Kalman Filter for 3D Object Tracking. *arXiv preprint arXiv:2505.12340* (2025).
- [24] Jirong Zha, Nan Zhou, Zhenyu Liu, Tao Sun, and Xinlei Chen. 2024. Diffusion-based filter for fast and accurate collaborative tracking with low data transmission. *Authorea Preprints* (2024).
- [25] Xiao Zhang, Feng Ding, and Erfu Yang. 2019. State estimation for bilinear systems through minimizing the covariance matrix of the state estimation errors. *International Journal of Adaptive Control and Signal Processing* 33, 7 (2019), 1157–1173.
- [26] Mattia Zorzi. 2016. Robust Kalman filtering under model perturbations. *IEEE Trans. Automat. Control* 62, 6 (2016), 2902–2907.